Dual-Axis Out-of-Distribution Detection

Dataset Preparation Summary

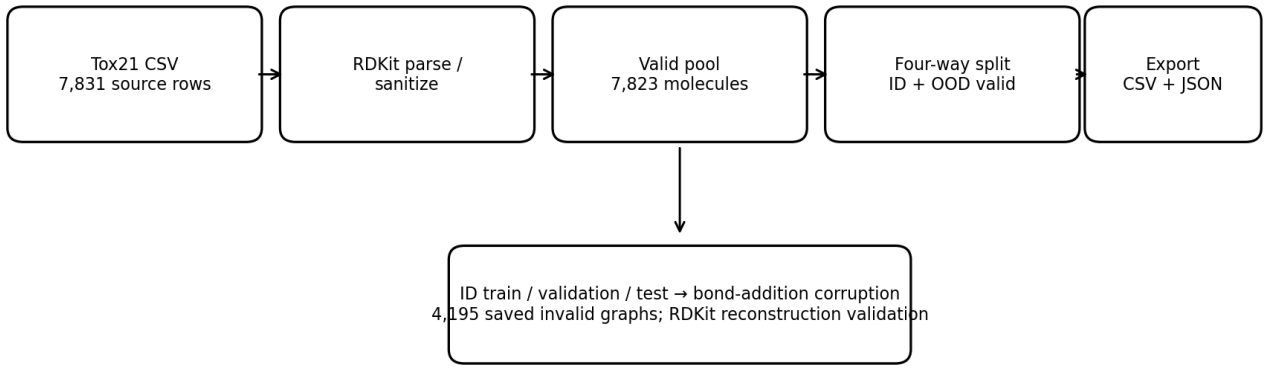
Tox21 • Dataset & Corruption Pipeline • Member 2

This report documents Member 2's dataset and corruption-pipeline work for a project that separates structural validity from semantic novelty. The prepared data uses Tox21 valid molecules for in-distribution (ID) training/validation/testing and a scaffold-held-out set of valid molecules for OOD evaluation. Deliberately corrupted graph records are generated separately to support structural-validity experiments.

1. Executive summary

- Loaded 7,831 source rows from Tox21.
- RDKit accepted 7,823 molecules; 8 source records failed parsing/sanitization and were audited separately.
- Created ID train, ID validation, ID test, and scaffold-held-out OOD-valid splits.
- Attempted corruption on 7,041 ID molecules; saved 4,195 invalid graphs (59.58% success rate).
- All 4,195 saved invalid graphs failed sanitization after reconstruction; reconstruction errors: 0.
- Export includes datasets, audits, validation reports, pilot outputs, and manifest.

2. Dataset preparation pipeline

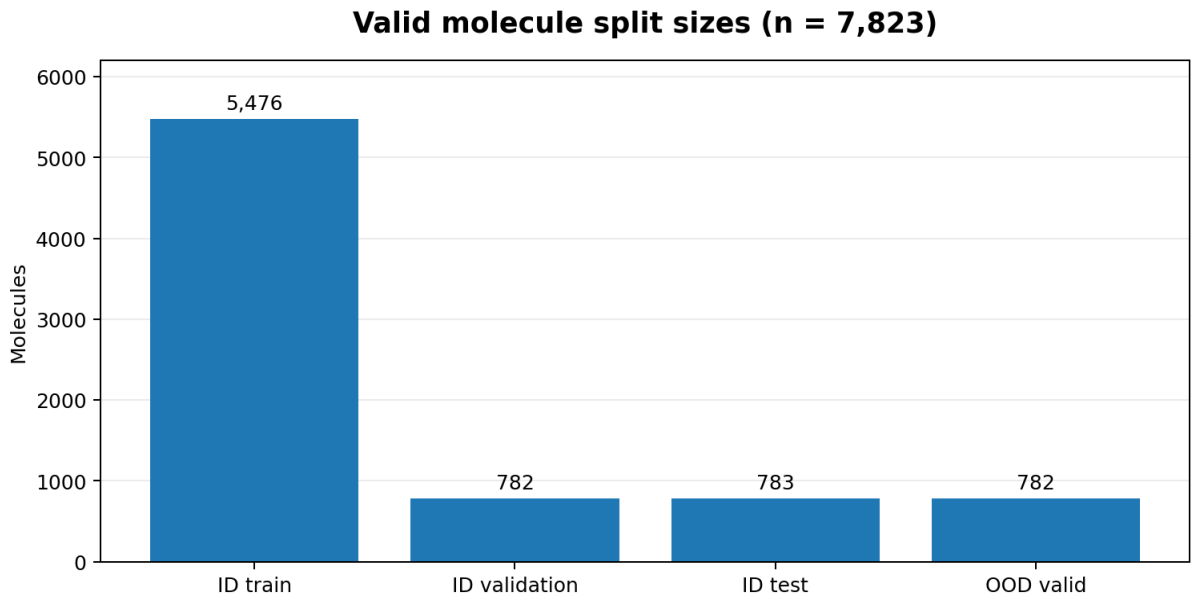


Source-invalid records are separate from generated invalid graphs. The latter are stored as JSON graph records, not valid SMILES.

3. Valid-molecule split strategy

The valid pool contains 7,823 unique canonical structures. The OOD-valid set was selected by whole scaffold groups (target 782). The remaining ID pool was randomly split into train/validation/test with seed 42; ID splits may share scaffolds by design.

Split	Molecules	Share of valid pool	Purpose
ID train	5,476	70.00%	Model training
ID validation	782	10.00%	Model selection / tuning
ID test	783	10.01%	Held-out ID evaluation
OOD valid	782	10.00%	Valid but scaffold-held-out evaluation
Total	7,823	100%	Accepted RDKit-valid molecules



4. Invalid graph generation

The v3 operator selects a neutral, non-aromatic carbon with total valence 4 and adds a single bond to a non-bonded neutral, non-aromatic carbon. A candidate is retained only if the modified carbon's explicit valence exceeds 4 and RDKit sanitization fails. Graphs are serialized as JSON.

Split	Attempted	Successful	Failed	Success rate
Train	5,476	3,237	2,239	59.11%
Validation	782	481	301	61.51%
Test	783	477	306	60.92%
Total	7,041	4,195	2,846	59.58%

The 2,846 unsuccessful attempts are not invalid graph examples; they are documented in the audit CSVs.

5. Reconstruction validation

- 4,195 / 4,195 reconstructed graphs failed RDKit sanitization (100%).
- Reconstruction errors: 0.
- This supports the structural-invalidity label for the saved graphs under the current validation procedure; it does not measure semantic novelty or model performance.

6. Output files

Category	Files
Valid molecule splits (4)	id_train.csv; id_val.csv; id_test.csv; ood_valid.csv
Corrupted graph datasets (3)	invalid_graphs_train.csv; invalid_graphs_val.csv; invalid_graphs_test.csv
Corruption audits (3)	corruption_audit_train.csv; corruption_audit_val.csv; corruption_audit_test.csv
Split IDs (1)	split_ids.csv
Source invalid audit (1)	source_invalid_audit.csv
RDKit validation reports (2)	rdkit_reconstruction_validation.csv; rdkit_validation_summary.csv

Category	Files
Manifest (1)	dataset_manifest.json
Pilot reports (3)	corruption_pilot.csv; corruption_pilot_v2.csv; corruption_pilot_v3.csv

7. Limitations and next steps

- Acyclic molecules use canonical SMILES as the fallback scaffold key; for these molecules, novelty is exact-structure based rather than scaffold-family based.
- Scaffold and canonical-SMILES non-overlap between ID and OOD were reported as checked in the split workflow; retain a final leakage-audit artifact for the handoff.
- Use valid ID and generated invalid graphs for structural-validity work; use valid OOD molecules for semantic-novelty evaluation.
- Keep the ZIP backup, manifest, and exact Colab notebook together for reproducibility.

Prepared from the reported Colab pipeline outputs. Counts and claims reflect those outputs; model performance is not evaluated in this report.